

Salvatore D. Pace

Curriculum Vitae

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Academic Positions

Institute for Advanced Study	Princeton, NJ
Member, School of Natural Sciences	2026 – Present

Education

Massachusetts Institute of Technology	Cambridge, MA
Ph.D. in Physics	2026
Thesis: <i>When Symmetry Does the Unexpected</i>	
Advisor: Xiao-Gang Wen	
University of Cambridge	Cambridge, UK
MPhil in Physics	2021
Thesis: <i>Emergent Axions in $U(1)$ Quantum Spin Liquids</i>	
Advisor: Claudio Castelnovo	
Boston University	Boston, MA
B.A. with honors & M.A. in Physics	2020
Thesis: <i>The Fine Structure Constant in Quantum Spin Ice</i>	
Advisor: Christopher Laumann	

Selected Awards and Honors

MIT Andrew M. Lockett Award for Doctoral Theoretical Physics Research	2026
Kavli Institute for Theoretical Physics Graduate Fellowship	2025
National Science Foundation Graduate Research Fellowship	2022 – 2025
Churchill Scholarship	2020 – 2021
American Physical Society LeRoy Apker Award Finalist	2020
BU College Prize for Excellence in the Physics Department	2020
Learning Assistant of the Year	2019
Goldwater Scholarship	2019

Scientific Papers [[Google Scholar](#)] [[INSPIRE](#)]

- [22] L. Gioia, [S. D. Pace](#), R. Verresen, S.-H. Shao, and R. Thorngren, “Disentangling the Toric Code,” [arXiv:2609.30363](#).
- [21] L. Z. Brito and [S. D. Pace](#), “Lattice 2-group symmetries: operators, defects, and gauging,” [arXiv:2609.10682](#).
- [20] E. Lew-Smith, [S. D. Pace](#) and S.-H. Shao, “Infinite-Order Lattice Chiral Anomalies and CPT,” [arXiv:2606.12510](#).
- [19] [S. D. Pace](#) and D. Bulmash, “Lieb-Schultz-Mattis constraints from stratified anomalies of modulated symmetries,” [arXiv:2602.11266](#).

- [18] M. L. Kim, S. D. Pace, and S.-H. Shao, “Symmetry-enforced Fermi surfaces,” *Phys. Rev. Lett.* **136** (2026) 176502, [[arXiv:2512.04150](#)].
- [17] S. D. Pace, Ö. M. Aksoy, and H. T. Lam, “Spacetime symmetry-enriched SymTFT: from LSM anomalies to modulated symmetries and beyond,” *SciPost Phys.* **20** (2026) 007, [[arXiv:2507.02036](#)].
- [16] S. D. Pace, M. L. Kim, A. Chatterjee, and S.-H. Shao, ‘Parity Anomaly from a Lieb-Schultz-Mattis Theorem: Exact Valley Symmetries on the Lattice,” *Phys. Rev. Lett.* **135** (2025) 236501, [[arXiv:2505.04684](#)].
- [15] S. D. Pace, A. Chatterjee, and S.-H. Shao, “Lattice T-duality from non-invertible symmetries in quantum spin chains,” *SciPost Phys.* **18** (2025) 121, [[arXiv:2412.18606](#)].
- [14] S. D. Pace, H. T. Lam, and Ö. M. Aksoy, “(SPT-)LSM theorems from projective non-invertible symmetries,” *SciPost Phys.* **18** (2025) 028, [[arXiv:2409.18113](#)].
- [13] A. Chatterjee, S. D. Pace, and S.-H. Shao, “Quantized axial charge of staggered fermions and the chiral anomaly,” *Phys. Rev. Lett.* **134** (2025) 021601, [[arXiv:2409.12220](#)].
- [12] S. D. Pace, G. Delfino, H. T. Lam, and Ö. M. Aksoy, “Gauging modulated symmetries: Kramers-Wannier dualities and non-invertible reflections,” *SciPost Phys.* **18** (2025) 021, [[arXiv:2406.12962](#)].
- [11] S. D. Pace and Y. L. Liu, “Topological aspects of brane fields: Solitons and higher-form symmetries,” *SciPost Phys.* **16** (2024) 128, [[arXiv:2311.09293](#)].
- [10] S. D. Pace, C. Zhu, A. Beaudry, and X.-G. Wen, “Generalized symmetries in singularity-free nonlinear σ models and their disordered phases,” *Phys. Rev. B* **110** (2024) 195149, [[arXiv:2310.08554](#)].
- [9] S. D. Pace, “Emergent generalized symmetries in ordered phases and applications to quantum disordering,” *SciPost Phys.* **17** (2024) 080, [[arXiv:2308.05730](#)].
- [8] S. D. Pace and X.-G. Wen, “Exact emergent higher-form symmetries in bosonic lattice models,” *Phys. Rev. B* **108** (2023) 195147, [[arXiv:2301.05261](#)].
- [7] Y.-T. Oh, S. D. Pace, J. H. Han, Y. You, and H.-Y. Lee, “Aspects of $\mathbb{Z}_N$ rank-2 gauge theory in $(2+1)$ dimensions: Construction schemes, holonomies, and sublattice one-form symmetries,” *Phys. Rev. B* **107** (2023) 155151, [[arXiv:2301.04706](#)].
- [6] S. D. Pace and X.-G. Wen, “Emergent higher-symmetry protected topological orders in the confined phase of $U(1)$ gauge theory,” *Phys. Rev. B* **107** (2023) 075112, [[arXiv:2207.03544](#)].
- [5] S. D. Pace and X.-G. Wen, “Position-dependent excitations and UV/IR mixing in the $\mathbb{Z}_N$ rank-2 toric code and its low-energy effective field theory,” *Phys. Rev. B* **106** (2022) 045145, [[arXiv:2204.07111](#)].
- [4] S. D. Pace, C. Castelnovo, and C. R. Laumann, “Dynamical Axions in $U(1)$ Quantum Spin Liquids,” *Phys. Rev. Lett.* **130** (2023) 076701, [[arXiv:2109.06890](#)].
- [3] S. D. Pace, S. C. Morampudi, R. Moessner, and C. R. Laumann, “Emergent Fine Structure Constant of Quantum Spin Ice Is Large,” *Phys. Rev. Lett.* **127** (2021) 117205, [[arXiv:2009.04499](#)].
- [2] S. D. Pace, K. A. Reiss, and D. K. Campbell, “The β Fermi-Pasta-Ulam-Tsingou Recurrence Problem,” *Chaos* **29** (2019) 113107, [[arXiv:1908.00564](#)].
- [1] S. D. Pace and D. K. Campbell, “Behavior and breakdown of higher-order Fermi-Pasta-Ulam-Tsingou recurrences,” *Chaos* **29** (2019) 023132, [[arXiv:1811.00663](#)].

Invited Talks

Colloquia

University of Minnesota [[Slides](#)]

Apr '26

Seminars

City College of New York [<i>slides to appear</i>]	Nov '26
Bard College [<i>slides to appear</i>]	Oct '26
University of Pennsylvania [<i>slides to appear</i>]	Oct '26
Institute for Advanced Study [Slides], [Recording]	Sept '26
University of Minnesota [Slides]	May '26
Institute for Advanced Study [Notes]	Jan '26
University of Oxford [Notes]	Nov '25
Simons Center for Geometry and Physics [Notes], [Recording]	Oct '25
CU Boulder CTQM [Slides]	Sept '25
Georgia Tech [Slides]	May '25
UCLA [Pre-talk notes], [Main talk slides]	Feb '25
Symmetry Seminar [Slides], [Recording]	Feb '25
Perimeter Institute for Theoretical Physics [Slides], [Recording]	Nov '24
Ohio State University [Slides]	Oct '24
Harvard [Slides]	Oct '24
Boston University [Notes]	May '24
Symmetry Seminar [Slides], [Recording]	Sept '23
Boston University [Slides]	June '22
Max Planck Institute for the Physics of Complex Systems [Slides]	Nov '20

Workshops/conferences

Leinweber workshop on Quantum Fields, Matter, and their Symmetries [<i>Slides to appear</i>]	Dec '26
IQIS Workshop: New Developments in Lattice Fermions [Slides]	Aug '26
Symmetries 26 [Slides], [Recording]	June '26
GGI Workshop: Defects and Extended Excitations in Quantum Field Theory, Quantum Matter and Statistical Models [Notes], [Recording]	June '26
Simons Collaboration Workshop on Confinement [Notes]	May '26
Aspen Conference: Generalized Symmetries and Defects in QFT and Gravity [Notes]	Feb '26
OIST TSVP Symposium: Aspects of Generalized Symmetries [Slides]	June '25
OIST Thematic Program: Generalized Symmetries in Quantum Matter [Pre-talk notes], [Main talk slides]	June '25
KITP Program: Generalized Symmetries in Quantum Field Theory: High Energy Physics, Condensed Matter, and Quantum Gravity [Slides], [Recording]	Apr '25
IBS PCS Workshop: Effective Field Theory Beyond Ordinary Symmetries [Slides], [Recording]	Dec '24
SCGP Workshop: Applications of Generalized Symmetries and Topological Defects to Quantum Matter [Slides], [Recording]	Sept '24

Teaching Experience

Schools and workshops

- Invited Lecturer: [2026 Arnold Sommerfeld School on Generalized Symmetries](#) Fall '26
[\[Lecture notes\]](#)
- Invited TA: [The Physics and Mathematics of Boundaries, Impurities, and Defects](#) Fall '25
[\[Lecture 1 recording\]](#), [\[Lecture 2 recording\]](#), [\[Notes\]](#)
- Invited TA: [Atlantic TQFT Spring School 2025](#) Spring '25

Massachusetts Institute of Technology

- Teaching Assistant
- 8.02, Physics II Spring '26
- Guest Lecturer
- 8.513, Modern Quantum Many-Body Physics Fall '23
- 8.231, Physics of Solids I Fall '22

Boston University

- Undergraduate Teaching Assistant
- PY406, Electromagnetic Fields and Waves II Spring '20
- PY405, Electromagnetic Fields and Waves I Fall '19
- PY452, Quantum Physics II Fall '19
- PY451, Quantum Physics I Spring '19
- PY410, Statistical Physics & Thermodynamics Spring '19
- PY351, Modern Physics I Fall '18
- PY313, Waves and Modern Physics Fall '18
- Guest Lecturer
- PY410, Statistical Physics & Thermodynamics Spring '19

Academic Services

Organizing

- [Symmetry seminar](#): Organizer 2025 – Present
- MIT Ultra Quantum Matter seminar: Organizer 2022 – 2026
- MIT physics colloquium committee: Graduate student representative 2021 – 2024

Mentorship

- Mentor for [Project SHORT](#) 2020 – Present
- MIT UROP Supervisor 2022 – 2023
- Mentor for [Boston University's PRISM](#) 2018 – 2020

Journal Referee: SciPost and Physical Review